

relearn

1 Learn

built-in analysis optimizer, an LLM agent

“Return amount in April?” + stated meaning
SELECT SUM(sr_return_amt) ...
WHERE d_moy = 4 AND ...
answer 3,133,115.63, judged correct
extracted, admission checks pass

**2 Publish revision 1**

what the definition records

SUM(sr_return_amt) by return date
no filter on store_returns
conditions: one row per return,
one date per date key, no lost dates
evidence: agent's SQL, table version 12

3 Benign write

late returns appended

store_returns 12 → 13
revision 1 pending
affected conditions: hold ✓
date_dim unchanged: kept ✓
still valid, still revision 1

✗ No unique fix

e.g. a full backup copy appended

sr_source='primary' ✓
sr_source='backup' ✓
both restore one row per return
invalidated, not repaired
relearned from a new judged task

a later write

4 Breaking write

an application-state row per return

store_returns 13 → 14
one return, two rows: completed, applied
one row per return fails: invalidated ✗
unmaintained answer for May:
6,588,699.86, not 3,294,349.93

5 Search filters

filters on columns with few values

sr_status='completed' ✓
one row per return, none lost
sr_status='applied' ✗
loses returns without an application row
exactly one fix: unique ✓

6 Regression test

both run on the data as of learning

revision 1 + sr_status='completed'
April, as of learning time:
agent's learning SQL 3,133,115.63
replacement 3,133,115.63 ✓
other admission checks pass ✓

7 Publish revision 2

same definition, new revision

revision 2, valid
filter sr_status='completed'
conditions re-derived
SQL declaring revision 1 rejected
waiting lookups get revision 2

two fixes, or none